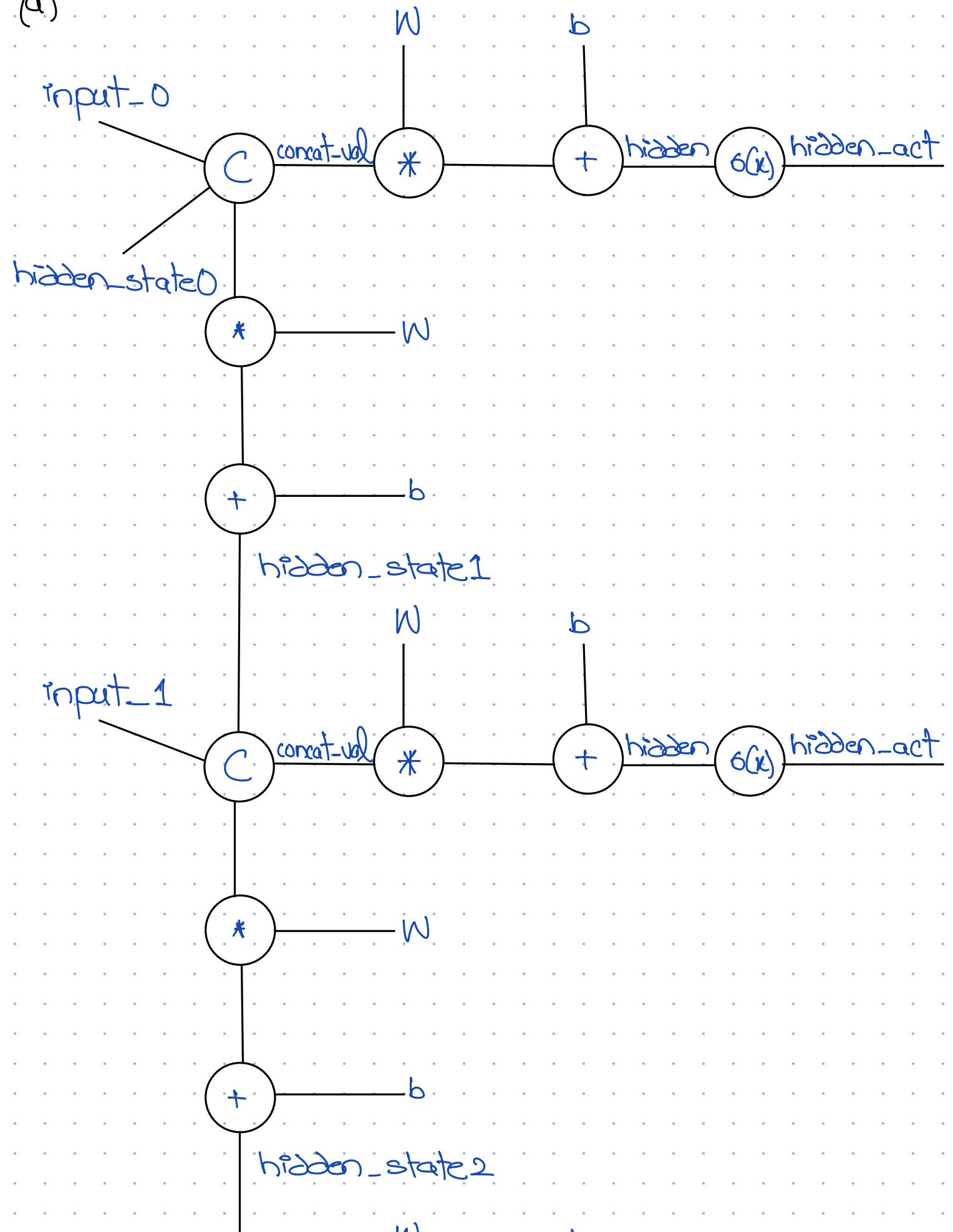
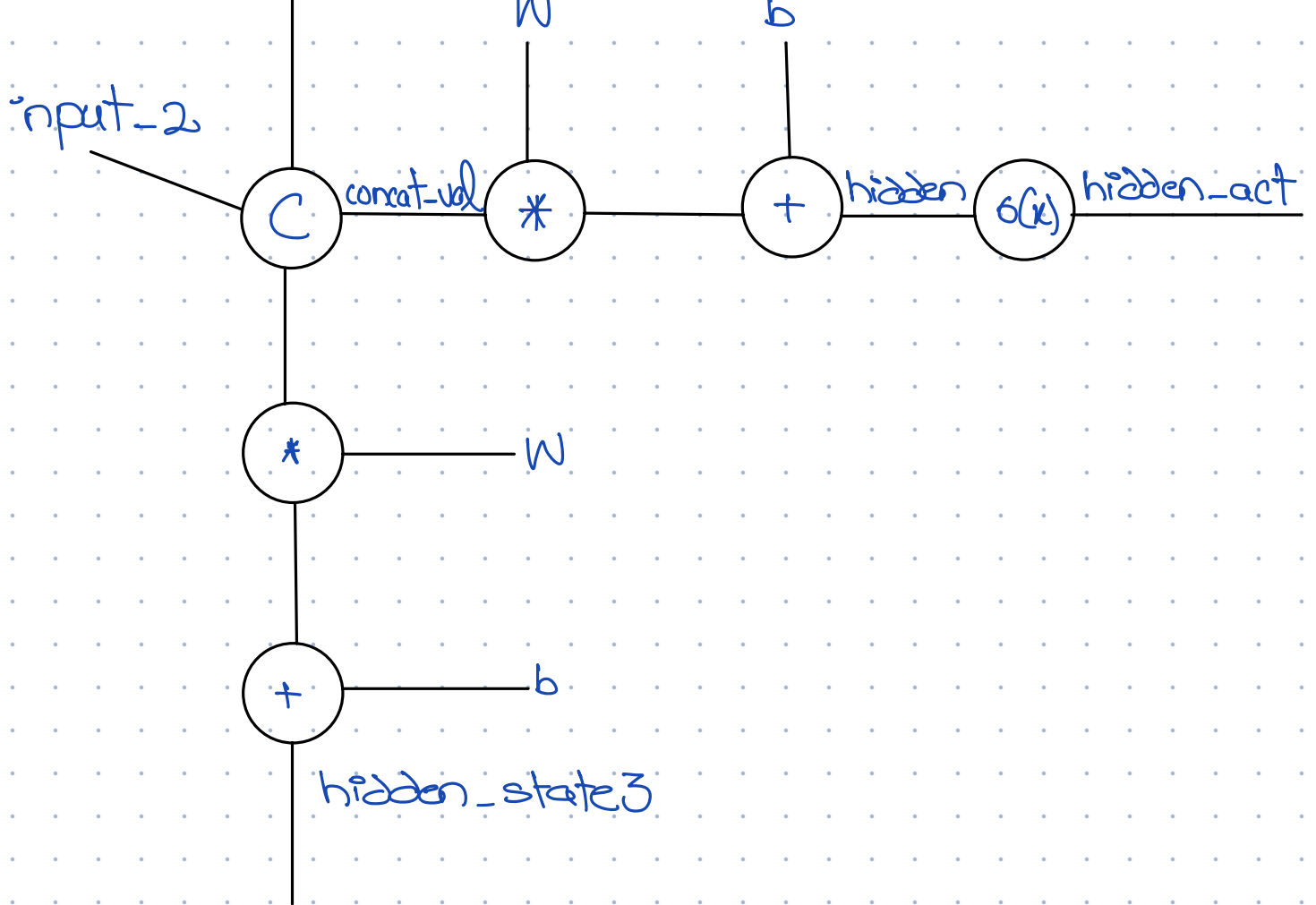


# Q1.2

(a)





Q1.3

|                      | 0   | 1   | 2    | 3   | 4   | 5    | 6   |
|----------------------|-----|-----|------|-----|-----|------|-----|
| $x_i$                | 0.0 | 1.0 | 2.0  | 3.0 | 4.0 | 5.0  | 6.0 |
| $\frac{dz}{dx}(x_i)$ | 2.0 | 1.5 | -2.5 | 0.5 | 1.0 | -1.0 | 0.0 |
| $z(x_i)$             | 5.5 | 7.5 | 9.0  | 6.5 | 7.0 | 8.0  | 7.0 |

$$\mathcal{E}(z) = \int_x \left( \frac{\partial z}{\partial x}(x_i) + p \right)^2 dx$$

$$\frac{dz}{dx}(x_i) = \frac{z(x_{i+1}) - z(x_i)}{x_{i+1} - x_i}$$

$$i=0,$$

$$\frac{dz(x_0)}{dx} = \frac{z(x_1) - z(x_0)}{x_1 - x_0}$$

$$2 = z(x_1) - z(x_0)$$

$$2 = 7.5 - z(x_0)$$

$$z(x_0) = 5.5$$

$$i=1,$$

$$\frac{dz(x_1)}{dx} = \frac{z(x_2) - z(x_1)}{x_2 - x_1}$$

$$1.5 = 9 - z(x_1)$$

$$z(x_1) = 9 - 1.5$$

$$= 7.5$$

$$i=2,$$

$$\frac{dz(x_2)}{dx} = \frac{z(x_3) - z(x_2)}{x_3 - x_2}$$

$$-2.5 = z(x_3) - z(x_2)$$

$$-2.5 = z(x_3) - 9$$

$$z(x_3) = 9 - 2.5$$

$$= 6.5$$

$$i=3,$$

$$\frac{dz(x_3)}{dx} = \frac{z(x_4) - z(x_3)}{x_4 - x_3}$$

$$0.5 = z(x_4) - z(x_3)$$

$$0.5 = z(x_4) - 6.5$$

$$z(x_4) = 6.5 + 0.5$$

$$= 7.0$$

$$i=4,$$

$$\frac{dz(x_4)}{dx} = \frac{z(x_5) - z(x_4)}{x_5 - x_4}$$

$$1 = z(x_5) - z(x_4)$$

$$1 = z(x_5) - 7$$

$$z(x_5) = 8$$

$$i=5,$$

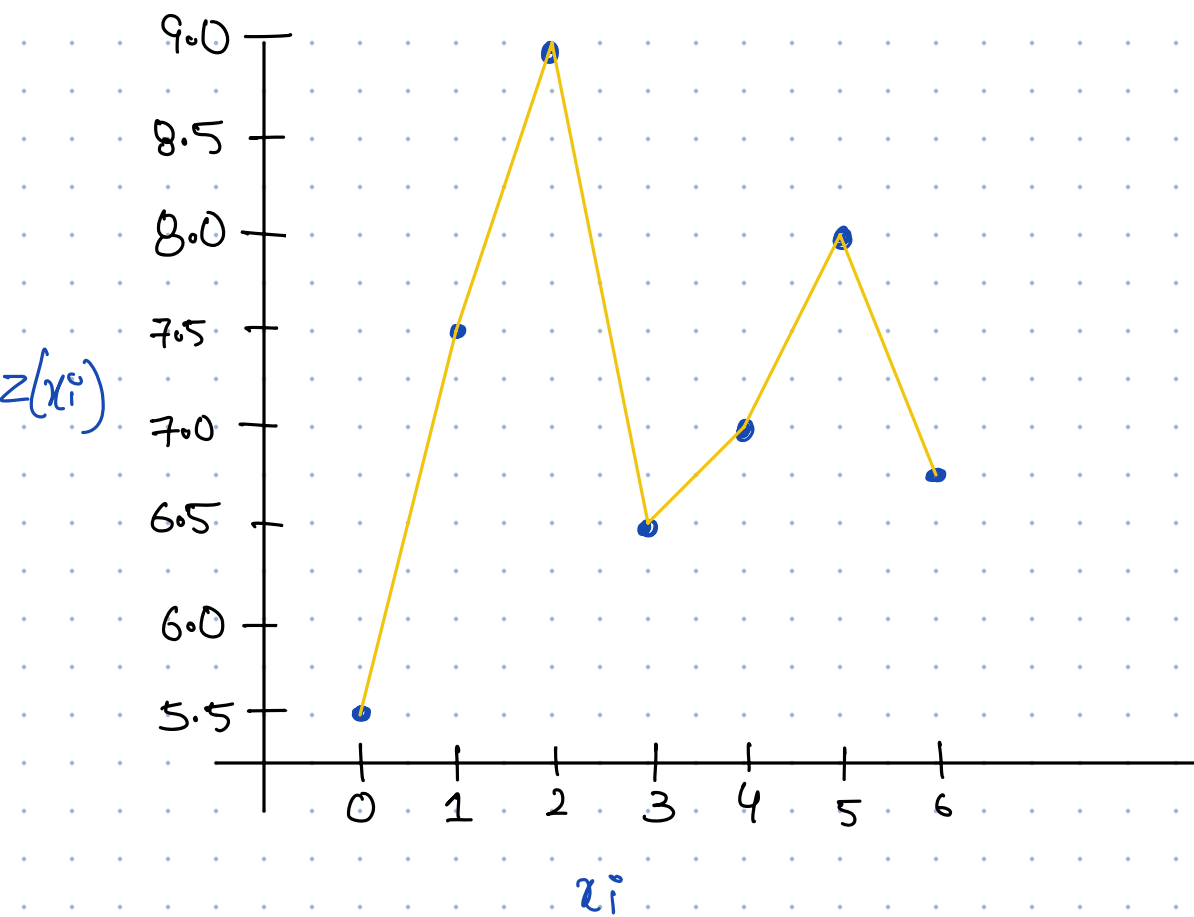
$$\frac{dz(x)}{dx} = \frac{z(x_6) - z(x_5)}{x_6 - x_5}$$

$$-1 = z(x_6) - z(x_5)$$

$$-1 = z(x_6) - 8$$

$$z(x_6) = 8 - 1$$

$$= 7$$



Q1.5:

$$(a) \quad s = [0.0, 0.0, 1.0]^T$$

$$R = 0.3, 0.65, 0.9$$

$$R(p, q) = r^T s$$

$$R(p, q) = \frac{ps_x + qsy + s_z}{\sqrt{p^2 + q^2 + 1}}$$

$$= \frac{1}{\sqrt{p^2 + q^2 + 1}}$$

$$= \frac{1}{\sqrt{p^2 + q^2 + 1}}$$

$$\sqrt{p^2 + q^2 + 1} = \frac{1}{R(p, q)}$$

$$p^2 + q^2 + 1 = \frac{1}{R(p, q)^2}$$

$$p^2 + q^2 = \frac{1}{R(p, q)^2} - 1$$

$$x^2 + y^2 = r^2$$

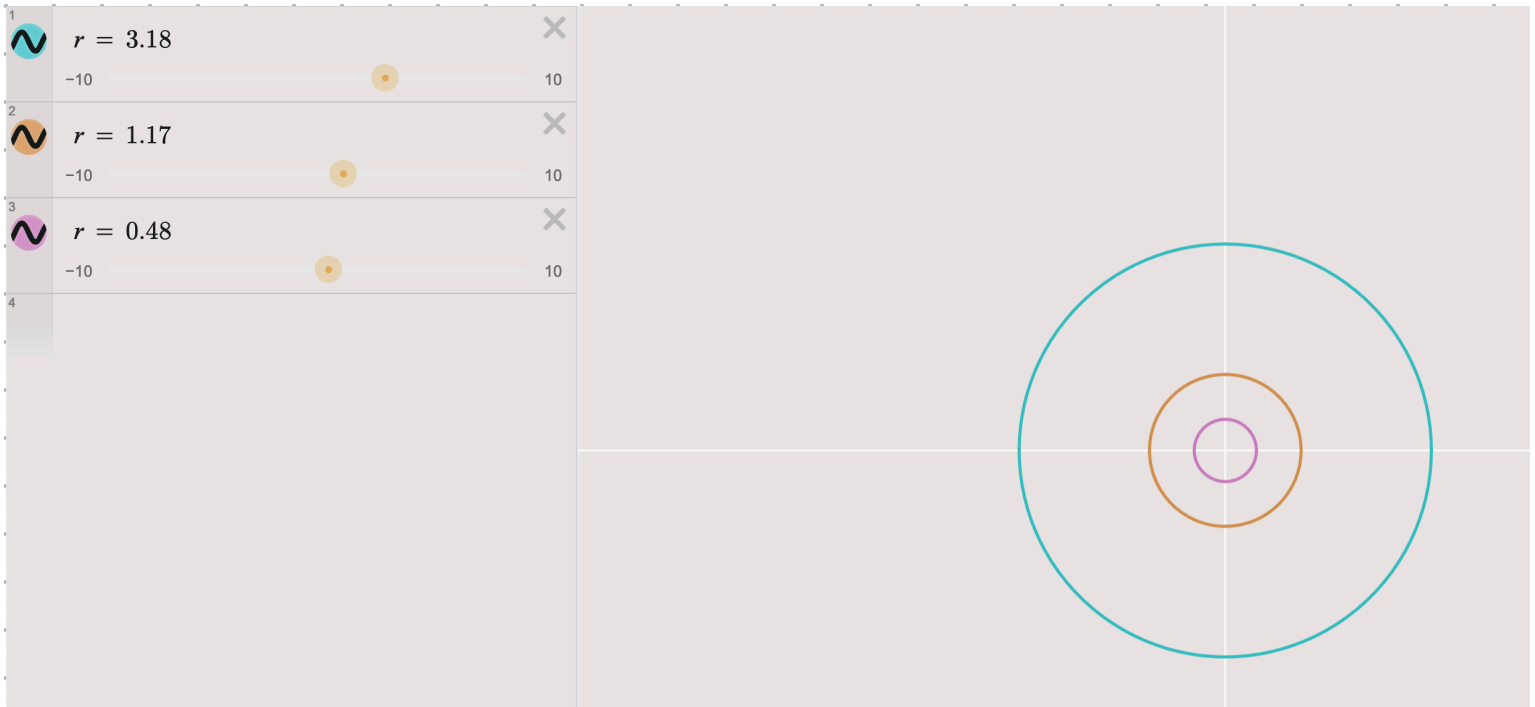
this is a circle equation centre  $(0, 0)$  <sup>a, b</sup>

$$\text{So, } r = \sqrt{\frac{1}{R(p, q)^2} - 1}$$

$$r = \sqrt{\frac{1}{(0.3)^2} - 1} = 3.18$$

$$r = \sqrt{\frac{1}{(0.65)^2} - 1} = 1.17$$

$$r = \sqrt{\frac{1}{(0.9)^2} - 1} = 0.48$$



(c)

$$f = \frac{2p}{1 + \sqrt{p^2 + q^2} + 1} \quad g = \frac{2q}{1 + \sqrt{p^2 + q^2} + 1}$$

$$f^2 + g^2 = \left( \frac{2p}{1 + \sqrt{p^2 + q^2} + 1} \right)^2 + \left( \frac{2q}{1 + \sqrt{p^2 + q^2} + 1} \right)^2$$

$$= \frac{4p^2}{\left(1 + \frac{1}{R}\right)^2} + \frac{4q^2}{\left(1 + \frac{1}{R}\right)^2}$$

$$= \frac{4}{\left(1 + \frac{1}{R}\right)^2} (p^2 + q^2)$$

$$= \underline{\underline{4 \left( \frac{1}{R^2} - 1 \right)}}$$

$$\begin{aligned}
 & \left(1 + \frac{1}{R}\right)^2 \\
 &= 4 \left[ \frac{1-R^2}{R^2} \times \left(\frac{R}{R+1}\right)^2 \right] \\
 &= 4 \left[ \frac{1-R^2}{(R+1)^2} \right]
 \end{aligned}$$

$$\gamma' = 2 \sqrt{\frac{1-R^2}{(R+1)^2}}$$

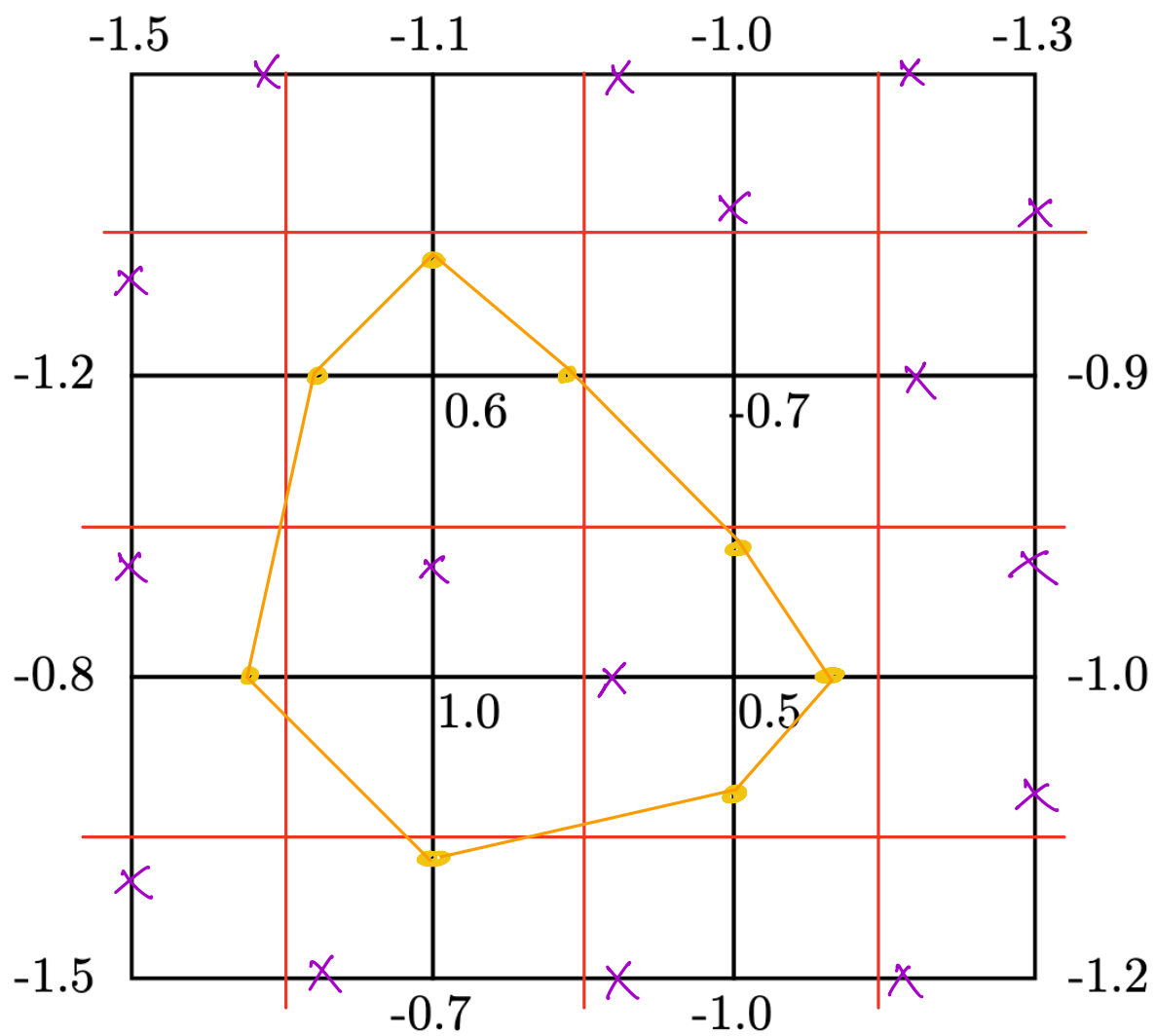
So,

$$\gamma' = 2 \sqrt{\frac{1-0.3}{(0.3+1)^2}} = 1.468$$

$$\gamma' = 2 \sqrt{\frac{1-0.9}{(0.9+1)^2}} = 0.45$$

$$\gamma' = 2 \sqrt{\frac{1-0.65}{(0.65+1)^2}} = 0.92$$

Q1.6





$$x = \frac{D_{i,j}}{D_{i,j} - D_{i,j+1}}$$

For (0,0):

$$x = \frac{-1.1}{-1.1 - 0.6}$$

$$= 0.64$$

For (1,0):

$$\frac{-1.2}{-1.2 - 0.6} = 0.66$$

Q1.1

$$\mathcal{L}(\mathbf{w}) = - \sum_{n=1}^N \left[ \psi(\mathcal{X}^n, \mathcal{Y}^n, \mathbf{w}) - \log \sum_{\mathcal{Y}} \exp \{ \psi(\mathcal{X}^n, \mathcal{Y}, \mathbf{w}) \} \right]$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = - \sum_{n=1}^N \left[ \nabla_{\mathbf{w}} \psi(\mathcal{X}^n, \mathcal{Y}^n, \mathbf{w}) - \nabla_{\mathbf{w}} \log \sum_{\mathcal{Y}} \exp \{ \psi(\mathcal{X}^n, \mathcal{Y}, \mathbf{w}) \} \right]$$

$$\nabla_{\mathbf{w}} \log \sum_{\mathcal{Y}} \exp \{ \psi(\mathcal{X}^n, \mathcal{Y}, \mathbf{w}) \} =$$

$$= \frac{1}{\sum_{\mathcal{Y}} \exp \{ \psi(\mathcal{X}^n, \mathcal{Y}, \mathbf{w}) \}} \sum_{\mathcal{Y}} \exp \{ \psi(\mathcal{X}^n, \mathcal{Y}, \mathbf{w}) \} \nabla_{\mathbf{w}} \psi(\mathcal{X}^n, \mathcal{Y}, \mathbf{w})$$

$$= \sum_{\gamma} \frac{\exp\{\psi(x^n, \gamma, \omega)\}}{\sum_{\gamma'} \exp\{\psi(x^n, \gamma', \omega)\}} \sigma_{\omega} \psi(x^n, \gamma, \omega)$$

$$= P(\gamma | x, \omega) \sigma_{\omega} \psi(x^n, \gamma, \omega)$$

So,

$$\frac{\partial L}{\partial \omega} = - \sum_{n=1}^N \left[ \nabla_{\omega} \psi(x^n, \gamma^n, \omega) - P(\gamma | x, \omega) \sigma_{\omega} \psi(x^n, \gamma, \omega) \right]$$

Q1.4

(Q)

$$\frac{\partial L}{\partial D} = \sum_i w_i (d_i - D)^2$$

$$0 = \sum_i 2w_i (d_i - D) (0 - 1)$$

$$0 = \sum_i -2w_i (d_i - D)$$

$$0 = -2 \sum_i w_i (d_i - D)$$

$$0 = \sum_i w_i d_i - \sum_i w_i D$$

$$\sum_i w_i D = \sum_i w_i d_i$$

$$D \sum_i w_i = \sum_i w_i d_i$$

$$D = \frac{\sum_i w_i d_i}{\sum_i w_i}$$

$$(b) \quad D(x) = \frac{1}{W(x)} \left[ w_i(x) d_i(x) + \sum_{n=1}^{i-1} w_n(x) d_n(x) \right]$$

$$\text{As, } w_i D(x) = \sum_i w_i(x) d_i(x)$$

$$\text{So, } D(x) = \frac{1}{w_i(x)} \left[ w_i(x) d_i(x) + w_{i-1}(x) D_{i-1}(x) \right]$$

Shift  $i$  by 1

$$\begin{aligned} D(x) &= \frac{1}{w_{i+1}(x)} \left[ w_i(x) D_i(x) + w_{i+1}(x) d_{i+1}(x) \right] \\ &= \frac{1}{w_i(x) + w_{i+1}(x)} \left[ w_i(x) D_i(x) + w_{i+1}(x) d_{i+1}(x) \right] \end{aligned}$$

$$\begin{aligned} w_{i+1}(x) &= \sum_i w_i(x) + w_{i+1}(x) \\ &= w_i(x) + w_{i+1}(x) \end{aligned}$$

# Coding

